

Original

Long-term effectiveness of arthrocentesis with and without hyaluronic acid injection for treatment of temporomandibular joint osteoarthritis

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Abstract: This study evaluated the long-term effectiveness of intra-articular temporomandibular joint arthrocentesis for patients with osteoarthritis and compared arthrocentesis/lavage alone with arthrocentesis/lavage and injected hyaluronic acid. Forty patients met the inclusion criteria, and 37 completed long-term follow-up (approximately 4 years). The patients were randomly allocated to two groups: arthrocentesis with lavage alone (A-group, $n = 17$) or combined with hyaluronic acid treatment (AS-group, $n = 20$). Standard two-needle arthrocentesis was performed. Pain and joint sounds were measured at baseline and approximately 4 years after treatment. Reported pain, as indicated by visual analogue scale (VAS) score, significantly decreased from baseline to the final follow-up examination in both groups. Mean VAS score decreased from 64 to 16 ($P < 0.001$) in the A-group and from 63 to 25 ($P < 0.001$) in the AS-group. Average maximum incisor opening increased significantly in both groups but did not significantly differ between groups ($P = 0.223$). Joint sounds did not significantly improve within groups (A-group, $P = 0.495$; AS-group, $P = 0.236$). Both methods resulted in significant long-term improvements in pain and jaw function.

Keywords: temporomandibular joint disorder; arthrocentesis; intra-articular injections; hyaluronic acid.

Introduction

Symptomatic derangement of the temporomandibular joint (TMJ) affects approximately 10% of the population, is more frequent in females, and increases in incidence with age (1,2). TMJ conditions may severely impair quality of life and can be challenging to treat (3). Osteoarthritis (OA) is a degenerative joint disease that destroys articular cartilage and bone, and is the most common condition affecting the TMJ (4). Individuals with OA of the TMJ experience restricted jaw movement, joint sounds (clicking, crepitation), and pain of varying intensity. Inflammation is the most important cause of TMJ-related pain, and substantial concentrations of inflammatory mediators such as interleukins (IL-1, IL-2, IL-6), tumor necrosis factor alpha, and cytotoxins have been found in the synovial fluid of patients with various types of arthritis (5,6). OA of the TMJ can be treated clinically with invasive procedures (arthroscopy, arthrocentesis, lavage, and hyaluronic acid [HA]) or corticosteroid injections) or noninvasive procedures (splints, massage, exercise, and medication).

Arthrocentesis is a simple, minimally invasive technique that flushes out inflammatory mediators from the superior joint space and disrupts adhesions within the joint. It was found to decrease intra-articular pain and joint sounds and increase mouth opening (7,8). Arthrocentesis can be performed under local anesthesia by means of standard two-needle technique and joint lavage with physiological solution or Ringer's solution. Further-

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more, arthrocentesis is much more easily tolerated than other surgical methods (e.g., open TMJ surgery) (9).

HA is a polysaccharide of low, medium, or high molecular weight and has been used successfully as a TMJ injection, to reduce inflammation and restore normal lubrication (4). Injected HA was significantly more effective than injected corticosteroids in decreasing pain intensity in OA of the TMJ (4). Similarly, Ungor et al. (10) concluded that HA injection in combination with arthrocentesis had promising effects on pain and TMJ function.

The short-term effects of intra-articular treatment with arthrocentesis have been evaluated in patients with OA (11,12) and disc displacement (13,14); however, few studies have investigated the long-term effects of combined arthrocentesis/lavage and HA injection in patients with TMJ disorders (10). No studies have examined the long-term effects of arthrocentesis and compared its effects with or without HA injection into the TMJ in patients with OA. This study therefore evaluated the long-term effects of arthrocentesis/lavage with and without injected HA in patients with OA of the TMJ.

Materials and Methods

Study site, randomization, and patients

The Regional Committees for Medical and Health Research Ethics of Norway (S-09323a) approved this randomized, blinded, prospective study. The participants were recruited from consecutive patients with a diagnosis of OA, as defined by the Research Diagnostic Criteria for Temporomandibular Disorders (RDC/TMD), (15) who were admitted to the Department of Oral Surgery and Oral Medicine of the University of Oslo. Inclusion criteria were OA in one or both TMJs, with pain or reduced jaw function, and radiographic evidence of OA, including erosions, flattening, and sclerosis of the condyle or articulating fossa. Panoramic radiographs were obtained for all 40 patients, and computed tomography (CT) and/or magnetic resonance (MR) images were additionally used if there was an indication (12 patients had Orthopantomogram (OPG) and CT images, 4 patients had OPG and MR images, and 3 patients had OPG, CT, and MR images). Exclusion criteria were a history of general arthritis or other connective tissue disease, immunosuppressive treatment, any organ disease, pregnancy or lactation, allergy or hypersensitivity to eggs, feathers, avian proteins, or chicken, or a history of intra-articular injection into the TMJ during the previous 12 months. Conservative therapies such as patient education, nonsteroidal anti-inflammatory drugs, physiotherapy, and occlusal splints were attempted in all

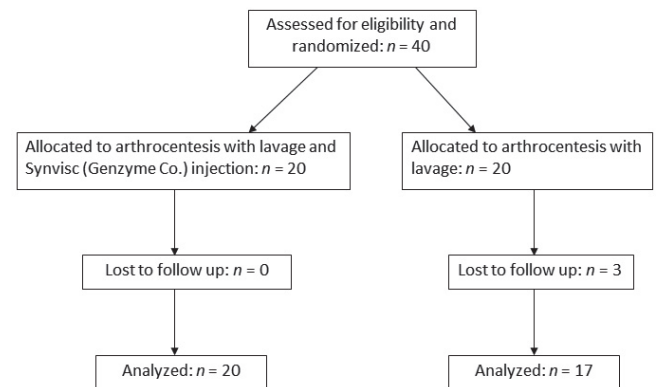


Fig. 1 CONSORT diagram showing flow of patients through the study.

patients before their enrollment in this study.

All patients provided written informed consent to participate in the study and were advised that they could withdraw at any time. If a patient had bilateral TMJ symptoms, the more severely affected side was studied, to avoid statistically complicated corrections of within-patient correlation. Forty patients were included in the study and treated with single side-arthrocentesis with lavage in the TMJ. The 40 patients were randomly allocated to two groups. One group received arthrocentesis with lavage, and the other group received arthrocentesis with lavage and an additional injection of HA (Synvisc, Genzyme Co., Ridgefield, NJ, USA; Fig. 1). All patients were instructed on how to maintain a soft-diet regimen for 14 days and were encouraged to practice self-assisted physical therapy for as long as needed.

Before the study started, 40 sealed envelopes were placed in a box: 20 contained a paper marked “AS”, for Synvisc (Genzyme Co.), and 20 contained a paper marked “A”, for arthrocentesis with lavage. The envelopes were mixed together. When a patient arrived for treatment, a sealed envelope was drawn from the box and opened to determine the treatment.

Of the 40 patients, 37 completed the study (Fig. 1). The mean duration of follow-up after treatment for these 37 patients was 47 months (range, 25-79 months). The baseline (pre-treatment) characteristics of the patients in each treatment group are shown in Table 1. Mean patient age was 51 years (range, 23-83 years).

Assessment of baseline jaw function included maximum incisor opening, lateral function on both sides, and protrusion of the lower jaw. Pain symptoms were evaluated at baseline with a visual analogue scale (VAS) from 0 to 100, where 0 was no pain and 100 was the worst imaginable pain. Pain localization was recorded as the TMJ or masticatory musculature. Sounds from the

Table 1 Baseline characteristics of patients who completed long-term follow-up ($n = 37$) after arthrocentesis with lavage and Synvisc (Genzyme Co.) injection (AS-group) or arthrocentesis with lavage alone (A-group).

	AS-group	A-group
patients (n)	20	17
females/males (n)	19/1	11/6*
mean age, years ($\pm$ SD**)	47 ± 15.7	55 ± 14.5
mean maximum incisor opening (mm)	34	37
mean VAS*** score (0-100)	63	64
mean duration of TMJ symptoms in months ($\pm$ SD)	64.2 ± 49.2	63.5 ± 72.6

*There were significantly more men in the A-group ($P < 0.02$).

** SD = standard deviation.

*** VAS = visual analogue scale.

affected jaw, such as clicking or crepitation, were noted. The contralateral TMJ was also examined for pain and jaw sounds. The mean VAS score and maximum opening at baseline are shown in Table 1.

Arthrocentesis procedure and follow-up

All 37 patients were treated with arthrocentesis and lavage, and 20 additionally received an injection of Synvisc (Genzyme Co.). After disinfection of the pre-auricular area, 1 mL of Carbocaine 30 mg/mL (Dentsply Ltd., Surrey, England) local anesthesia was injected through the skin to the TMJ capsule. Standard two-needle technique arthrocentesis was performed as previously described (9) with joint lavage with Ringer's solution (Fig. 2). An envelope was opened and, if so indicated, the participant received an approximately 1-mL intra-articular injection of Synvisc (Genzyme Co.) with fixed needles. After the procedure, patients were instructed to eat soft food, reduce jaw function for 1 week, and use paracetamol (up to 500 mg three times daily), if needed. The patients underwent regular postoperative follow-up during the first year. The final follow-up examination was performed at least 2 years postoperatively. This study was double-blinded: neither the patient nor the dentist performing long-term follow-up was aware of the patient's treatment status.

Statistical methods

Statistical analyses were performed with SPSS version 22 for Mac (IBM Co., New York, NY, USA). The paired t -test was used for continuous variables, and the Wilcoxon matched pairs test was used for ordinal scale variables. Between-group comparisons were done with the two-sample t -test, Mann-Whitney test, and Wilcoxon signed rank test (in accordance with the measuring scale of the analyzed variable). All tests were carried out with a



Fig. 2 Arthrocentesis and needle placement in the right TMJ of a 53-year-old woman with osteoarthritis.

confidence interval of 95%, and a P value of 0.05 or less was considered to indicate statistical significance.

Results

The groups did not significantly differ with respect to mean age, baseline maximum incisor opening, baseline VAS score, or mean duration of TMJ symptoms. The only significant difference was the number of men in the A-group (Table 1).

Jaw movement

There was an overall improvement in jaw movement from baseline to the final examination. Average maximum incisor opening increased significantly from baseline to the final examination in both groups, but the difference between groups was not significant ($P = 0.223$; Table 2). Maximum incisor opening at baseline and at the final examination are shown in Table 2. The mean increase from baseline to final examination was 5 mm ($P = 0.009$) in the AS-group and 6 mm ($P = 0.007$) in the A-group.

Analysis of lateral and contralateral excursion showed that jaw function from baseline to final examination increased in both groups but did not significantly differ within or between groups (Table 3). Protrusion slightly increased from baseline to final examination in the AS-group and slightly decreased in the A-group; however, the differences were not significant (Table 3). Interestingly, mean jaw protrusion increased from 6.0 mm at baseline to 7.2 mm at the 6-month follow-up examination in the AS-group and from 6.1 to 6.9 mm in the A-group.

Assessment of pain

In both the AS-group and A-group, reported pain significantly decreased from the pre-treatment examination to the final examination (Table 4). There were no significant

Table 2 Maximum incisor opening, in mm ($\pm$ SD), at baseline and the final examination (FFU).

treatment	baseline mean	baseline median	baseline range	FFU mean	FFU median	FFU range	change mean	P-value
AS-group (n = 20)	34 ($\pm$ 10)	34	17-50	39 ($\pm$ 8)	41	23-51	5 ($\pm$ 8)	0.009*
A-group (n = 17)	37 ($\pm$ 10)	35	25-57	43 ($\pm$ 8)	40	34-61	6 ($\pm$ 8)	0.007**
between group comparison								0.223***

* Significant increase in mean maximum incisor opening in AS-group from baseline to the FFU.

** Significant increase in mean maximum incisor opening in A-group from baseline to the FFU.

*** No significant difference in increase in mean maximum incisor opening between the AS-group and A-group.

Table 3 Lateral excursion, contralateral excursion, and protrusion values, in mm ($\pm$ SD), measured at baseline and the final examination (FFU) in the AS-group (arthrocentesis with lavage and Synvisc injection, Genzyme Co.) and A-group (arthrocentesis with lavage alone).

	AS-group			A-group				
	baseline mean	baseline range	FFU mean	FFU range	baseline mean	baseline range	FFU mean	FFU range
lateral excursion to affected side	7.6 ($\pm$ 3.3)	1-14	9.0 ($\pm$ 2.5)	5-14	7.5 ($\pm$ 3.5)	3-15	9.3 ($\pm$ 2.9)	4-14
contralateral excursion	7.2 ($\pm$ 2.7)	3-13	8.7 ($\pm$ 3.0)	3-14	6.7 ($\pm$ 2.7)	2-12	8.4 ($\pm$ 3.2)	2-14
protrusion	6.0 ($\pm$ 1.5)	4-9	6.1 ($\pm$ 2.8)	0-10	6.1 ($\pm$ 1.7)	3-9	5.8 ($\pm$ 2.4)	0-10

Table 4 Mean visual analogue scale (VAS), in mm where 0 indicates no pain and 100 the worst imaginable pain, pain localisation and joint sounds, measured at baseline (before treatment), at 6-months follow up and at the final follow (FFU) up in the AS-group (arthrocentesis with lavage and injection with Synvisc (Genzyme Co.) and in the A-group (arthrocentesis with lavage alone). Number of patients (n).

	AS-group			A-group		
	baseline	6 m	FFU	baseline	6 m	FFU
VAS ($\pm$ S.D)	63 ($\pm$ 16.6)	40 ($\pm$ 14.5)	25 ($\pm$ 27.4)*	64 ($\pm$ 24.8)	29 ($\pm$ 32.3)	16 ($\pm$ 20.5)**
Pain localisation (n)						
no pain	0	6	8	0	11	8
TMJ + musculature	15	7	5	12	6	8
TMJ	5	6	7	5	0	1
Joint sounds (n)						
no sound	6	9	7	6	11	6
clicking	2	1	2	1	0	5
crepitation	7	8	9	8	3	2
click+crepitation	5	1	2	2	3	4

* Significant decrease of VAS-score from before the treatment (baseline) to the final follow-up in the AS-group ($P < 0.001$).

** Significant decrease of VAS-score from before the treatment (baseline) to the final follow-up in the AS-group ($P < 0.001$).

differences between the groups. Mean VAS score at baseline was 63 in the AS-group and 64 in the A-group. Mean VAS score progressively decreased in both groups at each follow-up examination (Table 4). VAS score was significantly lower at the final examination than at baseline in both groups: 25 ($P < 0.001$) in the AS-group and 16 ($P < 0.001$) in the A-group. In the AS-group, 17 patients had a lower VAS score, one had the same VAS score (80), and two had higher VAS scores (5 and 43 points). VAS score in the AS-group ranged from 40 to 100 (median, 58.5; mean, 63.0) at baseline and from 0 to 94 (median, 17;

mean, 25.3) at the final examination. In the A-group, 15 patients had lower VAS scores, but two had higher VAS scores (4 and 3 points). VAS score in the A-group ranged from 15 to 100 (median, 70; mean, 63.7) at baseline and from 0 to 79 (median, 9; mean, 16.3) at the final examination. Change in VAS score did not significantly differ between groups ($P = 0.276$). Table 4 shows the results of the 6-month follow-up examination, for comparison of change in VAS score and other variables after a shorter duration of follow-up.

Pain localization

Pain in the TMJ and musculature significantly decreased from baseline to the final examination in both groups ($P = 0.008$; Table 4). No patient was free of pain at baseline (Table 4). At baseline, 15 patients in the AS-group had pain in the TMJ and musculature; at the final examination, four reported the same pain localization (TMJ and musculature), six had less pain in the TMJ only, and five had no pain. At baseline, five patients had pain in the TMJ only; at the final examination, one reported no change in pain localization (TMJ only), one had greater pain in the TMJ and musculature, and three had no pain. In the A-group, five patients had pain in the TMJ only at baseline; at the final examination, three had no pain, one had the same pain localization, and one had pain in the TMJ and musculature. At baseline, 12 patients in the A-group had pain in the TMJ and musculature; at the final examination, seven had no change in pain localization, and five were free of pain.

Joint sounds

There was no significant change in joint sounds from baseline to the final examination within groups (AS-group, $P = 0.236$; A-group, $P = 0.495$) or between groups ($P = 0.084$) (Table 4).

In the AS-group, one patient with no joint sounds at baseline developed clicking and crepitation, although VAS score decreased from 57 to 19 and maximum incisor opening increased from 32 to 36 mm. Of the 14 patients with clicking or crepitation, five had crepitation only at baseline but had crepitation and clicking at the final examination. Among the seven patients with clicking and crepitation at baseline, only three still had clicking and crepitation at the final examination, while one had clicking only, one had crepitation only, and one had no joint sounds.

Three patients in the A-group with no TMJ sounds at baseline developed clicking; the other three had no joint sounds after 2 or more years of follow-up. One patient with clicking in the A-group at baseline had clicking only at the final examination. Among the eight patients with clicking and crepitation at baseline, two had clicking and crepitation, three had crepitation only, and three had no joint sounds at the final examination. Among the two remaining patients with crepitation at baseline, one continued to have crepitation only and the other had clicking at the final examination.

Discussion

The results indicate that arthrocentesis reduced pain and increased jaw function after up to 4 years of follow-up.

This is consistent with a recent study of arthrocentesis alone for TMJ osteoarthritis (8). Sodium hyaluronate injection had no additional effect after 4 years of follow-up, which contrasts with findings from short-term studies (4,16).

As compared with baseline, mouth opening at the final examination was significantly increased in both groups, and a significant increase was already noted in the A-group at 6 months. Thus, arthrocentesis with lavage alone appears to be an adequate treatment for TMJ osteoarthritis. In a long-term study in Sweden (17), patients with painful disc displacement without reduction received anesthetics alone or anesthetics and arthrocentesis with lavage. Pain had decreased in both groups at a 3-year post-treatment follow-up examination, although the difference between groups was not significant. The difference in sex ratio between the present two groups, i.e., the lower female/male ratio in the A-group, may have affected the results in this present study. Therefore, future studies should ensure that the female/male ratio is equivalent. Apart from this study, only a few studies reported long-term follow-up of patients who received arthrocentesis. In a recent retrospective study, Nitzan et al. (8) concluded that about 80% of patients with TMJ OA who received arthrocentesis with lavage alone had favorable outcomes, namely, reduced pain and increased mouth opening, after a mean follow-up of 56 months. This present study confirms their results in patients who underwent arthrocentesis without additional sodium hyaluronate. Önder et al. (18) reported similar results in their study comparing patients with TMJ OA and a control group treated with occlusal splints and medication only. In the study group (which received arthrocentesis), peak maximum mouth opening was noted at 6 months rather than at the end of the mean long-term postoperative follow-up period of 36.7 months, although the differences were significant at both follow-up examinations.

It is unclear if different TMJ diagnoses can be compared in patients receiving arthrocentesis, but many studies have concluded that arthrocentesis has effects on various TMJ disorders. Studies of arthrocentesis for patients with disc derangement reported success rates of 83% and 84% (19,20). Arthrocentesis is widely used in the TMJ for patients with a diagnosis of disc derangement. Only a few long-term studies have evaluated arthrocentesis and sodium hyaluronate in patients with different degenerative conditions in the TMJ (10,17,18).

Sodium hyaluronate injection is widely used in orthopedic treatment, especially in the knee joint (21,22). The results of arthrocentesis with additional injections (such as corticosteroids, visco-supplementation, or platelet-

rich plasma) have varied greatly in patients with TMJ disorders, and no consensus has been reached regarding the most effective treatment (4,23-25). However, a recent study by Kilic and Güngörmüş (26) reported that arthrocentesis and platelet-rich plasma is not superior to arthrocentesis with HA injection and that arthrocentesis and HA injection is more acceptable for patients on a short-term basis. Several studies reported benefits for visco-supplementation after arthrocentesis (16,27-29). HA may have a beneficial effect on inflammatory mediators in OA (30), but it is unclear whether arthrocentesis alone or in combination with HA or corticosteroids yields the best results with respect to function and quality of life (31). As compared with arthrocentesis alone, combined use of HA for treatment of OA (4) may improve TMJ function more rapidly. However, improved joint function and motion may induce normal production of synovial fluid, independent of the medications used during arthrocentesis; thus, the difference between treatments may be limited.

No control group was included in this present study. There are very few studies of arthrocentesis with placebo or sham surgery in the TMJ. Therefore, additional studies comparing arthrocentesis with a control group receiving a placebo or sham surgery would be helpful. Sahlström et al. (7) and Baker et al. (17) reported positive results with only anesthesia and sham surgery (flushing saline in a cup next to the patient to simulate lavage) for patients who believed that additional surgery had been performed. In general orthopedics, sham surgery is more often used, and sham shoulder surgery may be as effective as labial repair of biceps tenodesis (32).

In the present study, five patients did not report a reduction in pain from baseline to the final examination. In the A-group, two patients had increased VAS scores, and in the AS-group, two patients had increased VAS scores and one patient had no change in VAS score. Three of these five patients had lower VAS scores at 6 months than at baseline or the final examination. This suggests that the effect varies among patients and that patients thus have differing needs for arthrocentesis retreatment to achieve a sustained effect. All five of these patients also had pain for 4 years or longer (range, 48-144 months) before treatment, which suggests they had more severe OA and required additional treatment.

A strength of this present study was that arthrocentesis was performed by just one operator and that the final examinations were performed by two independent researchers, who were unaware of treatment assignment, thus minimizing the risk of bias.

This present study started in 2013 and followed the

diagnostic criteria for osteoarthritis described by Dworkin et al. (15). Revised criteria were proposed in 2014 (2), and new criteria for osteoarthritis diagnosis might yield other treatment outcomes in the future.

Existing evidence indicates that arthrocentesis has a low risk of complications (24) and that the benefits greatly exceed the risks. This was confirmed in this present study: no patient developed complications other than transient paresis of the facial nerve due to the use of local anesthesia.

In conclusion, arthrocentesis of the TMJ reduced pain and improved function after short- and long-term observation. These outcomes were not changed by the use of medication during arthrocentesis.

Conflict of interest

The authors have no conflict of interest to declare.

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